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The Educational Effects of AI PengTalk on Korean Primary English Learners: A Two-Level Meta-Analysis

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Abstract

This study provides the first quantitative synthesis of the educational effects of 'AI PengTalk,' a government-launched AI speaking practice platform, on Korean primary English learners. Building on a prior qualitative synthesis of the tool's pedagogical mechanisms (Lee, 2026), this meta-analysis tested whether those mechanisms translate into measurable learning gains. A systematic search identified 16 quantitative studies. After influential-case diagnostics, 77 effect sizes (Hedges' *g*) nested in 28 independent samples (N = 928) were retained and analyzed using a two-level meta-analytic model. Effects on the cognitive domain were medium-sized (within-group $g = 0.68$; between-group $g = 0.53$), whereas affective effects were consistently smaller within each design ($g = 0.59$ and 0.41, respectively). Sixth graders showed a large cognitive gain ($g = 1.12$) relative to third through fifth graders ($g = 0.52$, $p = .047$), a pattern verifiable only in single-group designs, while teacher intervention and shorter implementation periods showed non-significant positive trends. Although the moderation evidence rests on a small number of single-group samples, these findings are directionally consistent with the 'Paradox of AI Scaffolding,' , and suggest that sustainable AI integration is likely to require teachers' hybrid scaffolding.

Key Words: AI chatbot, AI PengTalk, AI 펭톡, meta-analysis, effect size

I. Introduction

Artificial Intelligence (AI) technologies have rapidly permeated English language education worldwide, and conversational AI chatbots in particular have gained traction as a practical response to EFL learners' chronically limited access to authentic input and output opportunities (Huang et al., 2022; H. Lee & Lee, 2024; Zhang & Zou, 2022). South Korea offers an especially instructive case of this global trend. AI PengTalk, an English speaking-

practice application that the Ministry of Education built in partnership with the Educational Broadcasting System (EBS) and rolled out nationwide in 2021, was designed specifically for public primary education: it aligns its speaking tasks with the 2015 and 2022 Revised National English Curricula, provides real-time feedback on pronunciation through Automatic Speech Recognition (ASR), and lowers young learners' affective filter through gamification elements featuring the popular character 'Pengsoo.' Its nationwide deployment has generated a sizable body of empirical research within a relatively short period, making AI PengTalk one of the most intensively studied educational AI tools in the Korean primary English context.

In a previous study, this line of research was synthesized qualitatively. Drawing on 36 empirical studies, Lee (2026) conducted a thematic qualitative synthesis that identified how AI PengTalk is operationalized in authentic classrooms (i.e., blended and flipped learning, differentiated remedial instruction, and spaced practice), how its affordances are perceived differently across learner characteristics, and how teachers and peers mediate its technological and assessment constraints. Two findings from that synthesis are particularly consequential for the present study. First, the synthesis proposed the 'Paradox of AI Scaffolding': while the tool serves as a highly effective affective scaffold for lower-proficiency learners, affording them an evaluation-free 'psychological safety zone' (Lee, 2026), it simultaneously exhibits a ceiling effect and fading novelty for advanced learners, who quickly find its mechanical, evaluative feedback unchallenging. Second, the synthesis concluded that AI PengTalk delivers its greatest value when positioned within the regular curriculum as a 'cognitive rehearsal tool' rather than as a self-contained teaching agent, complemented by teachers' 'hybrid scaffolding,' in which data-informed instructional decisions are paired with warm, person-to-person emotional support (Lee, 2026).

However, a qualitative synthesis, by design, can identify pedagogical mechanisms but cannot quantify their consequences. It remains an open empirical question how large the academic and affective benefits of AI PengTalk actually are, and whether the learner- and implementation-related mechanisms uncovered qualitatively (learner level, teacher mediation, and the timing and duration of use) are reflected in statistically detectable variation in effect sizes. Indeed, the primary studies themselves report seemingly inconsistent quantitative outcomes. Single-group studies frequently report large pre-to-post gains in speaking performance and affective factors (e.g., Heo, 2024; Jang & Kim, 2022), whereas findings become mixed once comparison groups are involved: one study found a significant advantage in pronunciation but none in vocabulary (Rim & Han, 2024), and another observed speaking gains in both the treatment and comparison conditions, with affective outcomes largely unchanged (Y. Lee & Lee, 2024). Such inconsistency is precisely the condition under which a meta-analysis is most informative (Lee et al., 2019).

Therefore, as a direct follow-up to the qualitative synthesis, the present study conducted a two-level meta-analysis of the quantitative research on AI PengTalk in Korean primary English education. Methodologically, the study employed a multilevel effect-size approach in which multiple effect sizes are nested within independent samples (Lee et al., 2019; Lee et al., 2025; Van den Noortgate et al., 2013), thereby retaining the full richness of the reported outcomes, which span pronunciation, vocabulary, fluency, attitude, motivation, and self-efficacy, without violating the assumption of statistical independence. By doing so, this research seeks to provide an objective, evidence-based foundation for designing sustainable AI–human collaboration models in primary English classrooms, guided by the following two research questions:

RQ#1 (Main Effects): What is the magnitude of AI PengTalk's effects on primary English learners' cognitive and affective domains, estimated separately for single-group (within) and comparison-group (between) designs?

RQ#2 (Moderation): How do these effects vary as a function of (1) grade level, as a proxy for learners' cognitive maturity and accumulated proficiency, (2) teacher intervention, and (3) duration of use?

II. Literature Review

The review proceeds in three steps. It first outlines the theoretical and empirical background of AI chatbots in primary English education, then summarizes the qualitative evidence on AI PengTalk from which the present study's moderating variables were derived, and finally specifies the remaining gaps that the two research questions above are designed to address.

1. AI Chatbots in Primary English Education

The theoretical case for AI chatbots in EFL education rests on well-established cognitive and affective accounts of second language acquisition. Cognitively, chatbots create systematic opportunities for 'pushed output' (Swain, 1985) in environments where chances for authentic speaking are scarce, compelling learners to rehearse target vocabulary and syntax through repeated production. Recent evidence further suggests that these cognitive benefits are maximized when practice is distributed over expanding intervals rather than massed (Kang et al., 2014; Nakata, 2015). Affectively, practicing with a machine partner that passes no judgment eases the anxiety young learners feel when speaking and, in Krashen's (1982) terms, keeps the affective filter low, an effect that is amplified when learners develop a sense of social presence toward the AI character and come to treat it less as a device and more as a

social partner (Nass & Moon, 2000). Learners' affective dispositions also shape the emotional quality of chatbot interaction itself: in chatbot-assisted language learning, learners' willingness to communicate has been shown to influence their task emotions (H. Lee & Lee, 2026).

Empirical syntheses have generally supported these expectations, although their scope differs by region and context, which matters for interpreting the effect sizes they report. At the international level, Huang et al.'s (2022) systematic review of research on chatbots in language education documented consistent benefits across diverse countries and educational settings, although no overall effect size was pooled; a meta-analysis of robot-assisted language learning found a medium-sized average effect relative to non-robot conditions (d = 0.59; H. Lee & Lee, 2022), and a meta-analysis of AI-guided individualized language learning, which aggregated studies across countries and learner populations, found substantial within-group gains (d = 1.18) alongside a positive overall treatment effect relative to business-as-usual conditions (d = 0.39; H. Lee & Lee, 2024). Within the Korean EFL context specifically, a meta-analysis of chatbot-assisted English classes reported a mean effect size of 0.69 (J. Y. Lee & Hwang, 2022). More narrowly, in the Korean primary context, early studies applying commercial smart speakers (e.g., Google Assistant, Amazon Alexa) revealed novel conversational opportunities and reduced foreign language anxiety, but also critical limitations for young learners: difficulty recognizing Korean-accented English, linguistically demanding responses, and a lack of curriculum alignment (H. Kim et al., 2019; Shin, 2019). These constraints motivated a pivot toward pedagogically customized chatbots, of which AI PengTalk is the most prominent government-driven example.

2. Qualitative Evidence on AI PengTalk: Mechanisms and the Paradox of AI Scaffolding

Since its 2021 launch, empirical studies of AI PengTalk have proliferated, and Lee (2026) synthesized 36 of them thematically. The synthesis yielded three sets of findings. First, regarding operationalization, AI PengTalk is most prominently integrated through three instructional models: blended and flipped learning, in which the 'Topic World' function is assigned as pre-class home learning to reserve classroom time for interactive activities; differentiated remedial instruction, which offers individualized pacing in phonics and basic literacy for slower learners; and spaced practice, which schedules repeated speaking tasks at progressively longer gaps (for instance, after one, three, and then eight days) so as to offset the forgetting curve (M. Kim et al., 2023). The synthesis concluded that the tool works best when it functions as a 'cognitive rehearsal tool' within the regular curriculum rather than as a stand-alone instructor.

Second, regarding learner-level affordances, the synthesis identified the ‘Paradox of AI Scaffolding.’ For lower-proficiency learners, the tool operates as an effective affective scaffold: it removes peer evaluation pressure, permits unlimited retries within a psychological safety zone, and sustains motivation through gamified rewards (Heo, 2024; Woo & Kim, 2023) and through learners’ sense that the Pengsoo character is a peer-like conversational partner similar to themselves rather than an evaluating authority, which leads them to respond to the AI as a social actor (Nass & Moon, 2000; T. Yoon, 2024). For higher-proficiency learners, by contrast, the tool fails as a cognitive scaffold within the Zone of Proximal Development (Vygotsky, 1978): its automated assessment supplies simple evaluative feedback (e.g., ‘Excellent’) rather than the corrective feedback or meaning negotiation that advanced learners require (W. Choi, 2021; G. E. Jeong et al., 2021), so initial interest fades rapidly, a combination of a ceiling effect and fading novelty.

Third, regarding constraints and mediation, the synthesis documented recurrent ASR failures (particularly with fricatives and plosives) and overly rigid grading algorithms that frustrate learners whose legitimate utterances are marked incorrect. Successful classrooms resolved these contradictions through teachers’ multifaceted mediation (resolving technical glitches, supplying the pronunciation corrections the AI misses, and providing emotional support), supplemented by heterogeneous peer pairing. Collectively, these strategies were conceptualized as ‘hybrid scaffolding’: teachers draw on the learning data the platform generates while supplying the human responsiveness the technology itself cannot provide.

3. The Present Research Gap: From Qualitative Mechanisms to Quantitative Verification

While the qualitative synthesis clarified how AI PengTalk works, it could not establish how well it works. Three specific gaps remain. First, no study has statistically aggregated the effect sizes reported across the quantitative AI PengTalk literature; existing effectiveness claims rest on individual studies with small samples, often a single intact class (e.g., Heo, 2024; Jang & Kim, 2022), and heterogeneous outcome measures, which makes their apparent inconsistencies difficult to adjudicate. Second, the moderating mechanisms proposed qualitatively remain untested quantitatively. The qualitative synthesis (Lee, 2026) provides a principled basis for selecting moderators: the Paradox of AI Scaffolding implicates learner-level characteristics (operationalized here as grade level, a proxy for cognitive maturity and accumulated proficiency), the hybrid scaffolding account implicates teacher intervention, and the fading novelty account implicates duration of use. Whether and how these variables actually moderate the tool’s effects is an open empirical question that only a quantitative synthesis can address. Third, previous reviews and meta-analyses of chatbot-assisted learning (e.g., Huang et al., 2022; J. Y. Lee & Hwang, 2022; H. Lee & Lee, 2024) aggregated across

heterogeneous tools, learner ages, and contexts, so their estimates cannot be attributed to any specific curriculum-aligned tool for young learners. A tool-specific meta-analysis of AI PengTalk, a single system deployed under a single national curriculum to a well-defined learner population, therefore offers a rare single-tool test case in which tool- and curriculum-level heterogeneity is held constant by design. To be clear, this does not render the evidence base homogeneous: implementation-level differences in instructional models, teacher mediation, and duration remain, as do measurement-level differences across studies, and the variation they generate is precisely what the moderator analyses in this study are designed to probe.

Methodologically, the present study also responds to the accumulation of multiple, conceptually distinct outcomes within primary studies. A single AI PengTalk study frequently reports separate scores for pronunciation, vocabulary, and fluency, alongside attitude, confidence, and self-efficacy scales. The conventional practice of selecting or averaging one effect size per study discards this within-study information. Following Lee et al. (2019), this study instead retained multiple effect sizes per sample and analyzed them with a multilevel approach; the following section details the two-level specification and the rationale behind it.

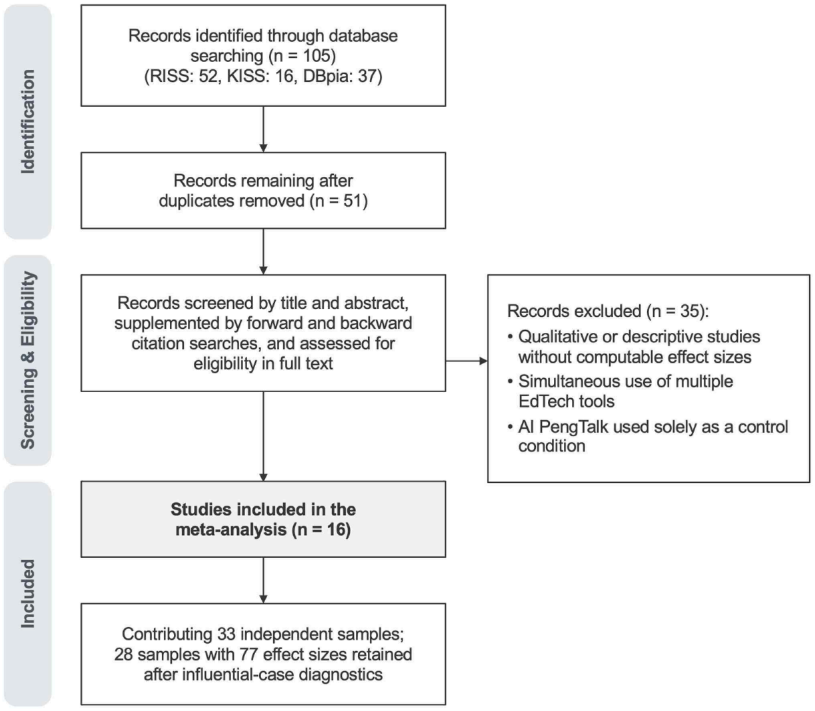
III. Methods

1. Literature Search and Study Selection

The literature search followed the same systematic procedure as the preceding qualitative synthesis (Lee, 2026) to ensure that the two studies draw on a common evidence base. Three major domestic academic databases (RISS, KISS, and DBpia) were searched with the keywords ‘AI 펭톡,’ ‘펭톡,’ ‘AI PengTalk,’ and ‘PengTalk,’ covering the period from the system’s public launch in 2021 to June 2026. The initial search yielded 105 records (RISS: 52, KISS: 16, DBpia: 37), of which 51 remained after duplicate removal. Titles and abstracts were screened, forward and backward citation searches were conducted, and the remaining records were subjected to full-text review.

For the present meta-analysis, studies were retained only if they permitted effect size computation. Specifically, a study was included if it (a) employed a quantitative experimental or quasi-experimental design targeting AI PengTalk in the Korean primary English context, and (b) reported sufficient statistical information (means, standard deviations, and sample sizes, or convertible inferential statistics such as t values) to calculate Hedges’ *g*. Studies were excluded if they (a) were qualitative or descriptive reports without analyzable quantitative data, (b) combined AI PengTalk with other educational technologies (such as metaverse platforms) within a single intervention, so that its unique contribution could not be

disentangled, or (c) treated AI PengTalk only as the comparison treatment rather than the focal intervention. Applying these criteria yielded a final pool of 16 primary studies (marked with asterisks in the reference list). Figure 1 illustrates the search and selection process.



<Figure 1> Flowchart of the Literature Search and Selection Process

2. Effect Size Calculation and Dataset Construction

All effect sizes were computed as Hedges' *g* (Hedges & Olkin, 1985), which corrects the small-sample bias of Cohen's *d*, an important consideration given that many primary studies involved single intact classes with fewer than 30 students. For single-group pretest–posttest designs, within-group effect sizes were computed from pretest and posttest means and standard deviations as the standardized mean change: the pretest–posttest mean difference was divided by the square root of the average of the squared pretest and posttest standard deviations, and the sampling variance was computed as $\mathrm{Var}(g) = (1/n + g^2/(2n)) \times 2(1 - r)$ (Borenstein et al., 2009). Because none of the primary studies reported the pretest–posttest correlation *r*, it was fixed at .50 for all within-group effect sizes, a conventional assumption in the absence of reported values. For two studies that reported only paired t statistics without descriptive statistics (M. Kim et al., 2023; K. Lee, 2022), the effect size was converted from *t* by treating the pretest and posttest as two independent groups of equal size, an approximation that yields larger, and thus more conservative, sampling variances than the repeated-measures formula under the assumed correlation. For comparison-group designs, between-group effect sizes were computed from posttest differences between the treatment and control groups,

standardized by the pooled standard deviation, with sampling variance $Var(g) = (n_1 + n_2)/(n_1n_2) + g^2/(2(n_1 + n_2))$. When a study permitted both within- and between-group comparisons, the two types of effect sizes were retained and analyzed separately, and within- and between-group estimates were never pooled into a single distribution (Lee et al., 2019).

Because primary studies typically reported multiple outcomes, such as separate subscores for pronunciation accuracy, intonation, and speech rate, or separate scales for interest, confidence, and attitude, multiple effect sizes were computed per sample rather than averaged. Each effect size was then classified by outcome domain: the cognitive domain comprised pronunciation, vocabulary, phonics and word recognition, and speaking fluency and accuracy measures, whereas the affective domain comprised attitude and interest, confidence and motivation, and self-efficacy measures. Crossing research design with outcome domain yielded four analytic strata: within-cognitive, within-affective, between-cognitive, and between-affective.

The 16 primary studies initially yielded 33 independent samples (N = 1,141). Following the convention of prior multilevel meta-analyses (Lee et al., 2019), an independent sample was defined as the group of learners contributing effect sizes to a given design and outcome domain, with within-group and between-group samples counted separately; the 33 samples thus comprised 14 within-cognitive, 9 within-affective, 6 between-cognitive, and 4 between-affective samples. Under this definition, a learner group providing both cognitive and affective outcomes, or both within- and between-group comparisons, contributes to more than one sample, so the summed N counts such learners once per sample; Appendix A documents this structure for each of the 16 studies.

Prior to the main analyses, influential-case diagnostics were conducted at the sample level within each design-by-domain stratum. Specifically, a random-effects model was fitted to each stratum using the metafor package in R (Viechtbauer, 2010), and the standard set of influence diagnostics was computed and inspected iteratively (Viechtbauer & Cheung, 2010). The diagnostics included studentized deleted residuals, DFFITS values, Cook’s distances, hat values, and leave-one-out estimates, with samples flagged as influential according to the cutoffs implemented in the package. Whenever a sample was flagged as influential, it was removed and the diagnostics were re-run on the reduced dataset until no influential cases remained. This procedure removed two samples from the within-cognitive stratum across two iterations (M. Kim, 2026; Heo, 2024) and three samples from the within-affective stratum across two iterations (Heo, 2024, in the first iteration; Cha, 2023, and J. W. Jeong, 2025, in the second); the removed samples are marked in Appendix A. In the between-group strata, the diagnostics were computed but case deletion was not applied: with only six cognitive and four affective samples, deletion-based diagnostics are unstable, and iterative removal would have eliminated most of the samples in these strata. All between-group samples were therefore

retained, and the between-group estimates should be interpreted with this caveat in mind. The final dataset comprised 77 effect sizes nested within 28 independent samples (N = 928): 12 within-cognitive (35 effect sizes), 6 within-affective (16 effect sizes), 6 between-cognitive (13 effect sizes), and 4 between-affective (13 effect sizes) samples. In addition, publication bias was assessed on the final analytic dataset at the level of independent samples, where the diagnostics' independence assumption is met: effect sizes were first aggregated to a single estimate per sample, and funnel plots were inspected and regression-based Egger tests under the random-effects model were conducted (Egger et al., 1997), separately for the within-group and between-group samples; the results are reported at the beginning of the Results section.

3. Coding of Moderator Variables

Guided by the mechanisms identified in the qualitative synthesis, three moderator variables were coded for each independent sample. First, grade level was coded from each study's participant description (Grades 3 through 6, the grades in which English is formally taught in Korean primary schools); for the moderator analysis, samples were contrasted as sixth grade versus Grades 3–5, treating grade as a proxy for learners' cognitive maturity and accumulated English proficiency. Two considerations motivated this contrast rather than the curriculum's grade-band division (Grades 3–4 vs. Grades 5–6). Theoretically, the mechanism at issue, the ceiling effect proposed for the most advanced learners in the Paradox of AI Scaffolding, concerns the upper end of the primary years, and sixth grade, the final primary year with the greatest cognitive maturity and the longest accumulated exposure to English instruction, isolates this upper end most directly. Practically, the distribution of the analyzable samples argued against the grade-band contrast: only two of the 11 within-cognitive samples entering the moderator analysis were drawn from Grades 3–4, so a Grades 3–4 versus Grades 5–6 comparison would have rested on a severely unbalanced split of two versus nine samples, whereas the sixth-grade contrast yields a more balanced split of four versus seven. Second, teacher intervention was coded dichotomously: implementations in which the teacher actively structured, monitored, or supplemented AI PengTalk use during instruction (e.g., pre- and post-practice activities orchestrated around the app) were coded as involving teacher intervention, whereas purely self-directed uses (e.g., independent home learning assignments without instructional mediation) were coded as not involving it. Third, duration of use was coded dichotomously based on each study's reported treatment schedule, contrasting semester-long implementations with shorter ones. The coding procedure was identical to that of the preceding qualitative synthesis (Lee, 2026): two researchers, including the author, independently coded a subset of the studies to develop and calibrate the coding scheme,

reaching complete (100%) initial agreement; this level of agreement reflects the low-inference nature of the coded variables, which were read directly from the primary studies' method sections (e.g., grade level, research design, and reported treatment schedules).

4. Data Analysis Plan

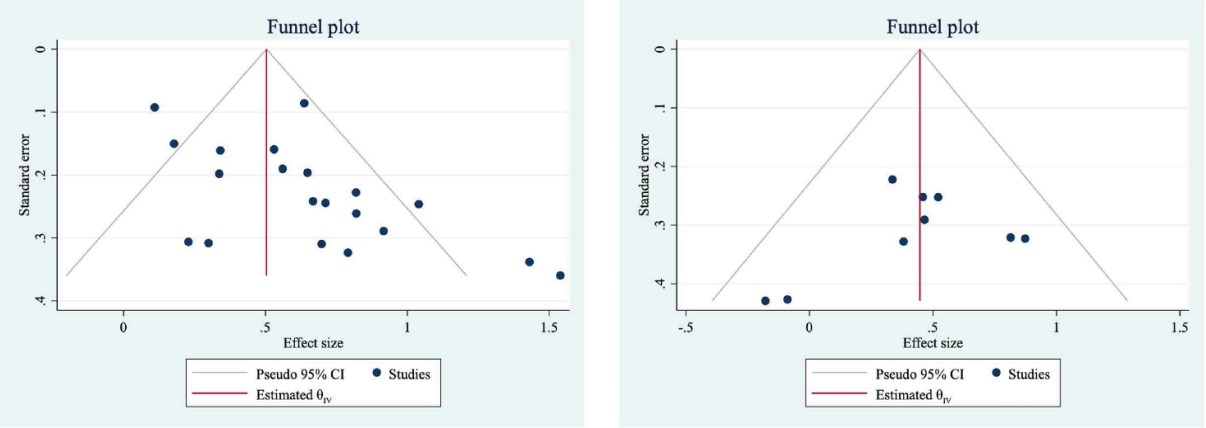
Given that multiple effect sizes were nested within independent samples, a two-level meta-analytic model was estimated, with effect sizes at Level 1 and independent samples at Level 2 (Hox et al., 2010; Van den Noortgate et al., 2013). This modeling choice is not merely technical: simply pooling all 77 effect sizes in a conventional analysis would violate the independence assumption, because effect sizes drawn from the same sample of students are more similar to one another than those drawn from different samples (Raudenbush & Bryk, 2002). The two-level model resolves this in three steps, which can be illustrated with the within-group data used for RQ#1. First, each of the 51 within-group effect sizes entered the model at Level 1 together with its known sampling variance, computed from the formulas above; the vocabulary effect size of Cha (2023), for instance, entered as $g = 0.92$ with a sampling variance of 0.08 based on its 17 learners. Second, in the variance-known specification (Lee et al., 2019), the sampling variance of each effect size was entered as a known quantity rather than estimated, so differences in sample size are controlled directly: effect sizes based on more learners carry smaller sampling variances. Beyond this known sampling error, the model estimates two variance components: a Level 1 residual variance, capturing heterogeneity among effect sizes within the same sample, and a Level 2 variance, capturing true heterogeneity between samples. Third, each effect size was weighted by the inverse of its total variance, the sum of its known sampling variance and the two estimated variance components, while effect sizes from the same sample shared a common sample-level random effect, which prevents a sample reporting many outcomes from dominating the pooled estimate. The Level 2 clustering unit was the independent sample as defined in the preceding section, so that the cognitive and affective effect sizes of the same learner group entered the model as separate samples. This specification captures the dominant source of dependence in the data, that is, multiple outcome measures of the same domain administered to the same learners; the residual dependence between a learner group's cognitive and affective effect sizes is not explicitly modeled. Because the domain-specific mean estimates for RQ#1 are each derived from their own domain's effect sizes, this simplification primarily affects the precision of the cognitive–affective contrast rather than the domain-specific estimates themselves, and it is acknowledged as a limitation in the concluding section. All models were estimated in Stata 16.1 using the meglm command.

For RQ#1, a two-level model including a domain indicator (cognitive vs. affective) was estimated separately for the within-group and the between-group effect sizes, and adjusted mean estimates for the four design-by-domain strata were derived as predictive margins; because the influential samples identified in the cognitive and affective diagnostics did not overlap, all five were excluded from the combined within-group model. Moderator analyses (RQ#2) then added grade level (sixth grade vs. Grades 3–5), teacher intervention, and duration as simultaneous predictors. Because the between-group strata contained too few samples (six cognitive and four affective), no moderator models were estimated for them; moderator analyses were conducted only for the within-group strata, and this constraint is flagged explicitly when interpreting the results. Following conventional benchmarks (Cohen, 1988), effect sizes of 0.20, 0.50, and 0.80 were interpreted as small, medium, and large, respectively.

IV. Results

1. Overall Effects of AI PengTalk (RQ#1)

Before the main analyses, publication bias was assessed at the sample level. In the analytic dataset, that is, with the five influential within-group samples already excluded, the regression-based Egger tests indicated no significant small-study effects for either the within-group (k = 18, b_1 = 1.51, SE = 0.89, z = 1.69, p = .092) or the between-group samples (k = 10, b_1 = −1.61, SE = 2.32, z = −0.69, p = .489), and visual inspection of the funnel plots revealed no marked asymmetry (Figure 2).



<Figure 2> Funnel Plots for the Within-Group (left) and Between-Group (right) Samples

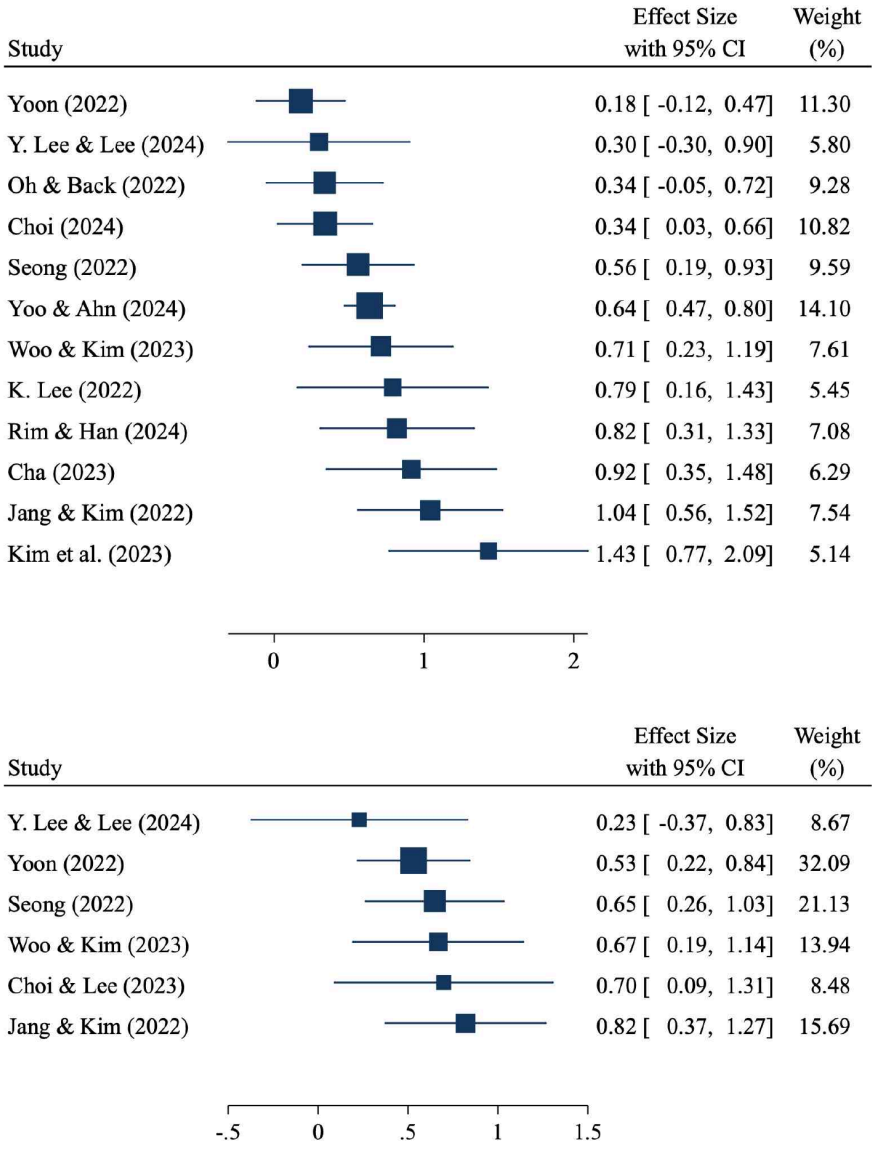
Table 1 presents the estimated mean effect sizes for the four design-by-domain strata. In the cognitive domain, AI PengTalk showed medium-sized positive effects under both designs: the single-group (within) estimate was 0.68 (SE = 0.10, 95% CI [0.47, 0.88], p < .001) and the comparison-group (between) estimate was 0.53 (SE = 0.17, 95% CI [0.20, 0.86], p = .002). In

the affective domain, the corresponding estimates were 0.59 (*SE* = 0.14, 95% CI [0.31, 0.86], $p < .001$) and 0.41 (*SE* = 0.17, 95% CI [0.08, 0.74], $p = .014$). In both models, the variance components indicated meaningful variation at both levels (within-group model: Level 2 [between-sample] variance = 0.06, Level 1 residual variance = 0.09; between-group model: 0.07 and 0.05, respectively), supporting the use of the two-level specification. Figures 3 and 4 display the distributions of the sample-level (aggregated) effect sizes underlying these estimates, by domain, for the within-group and between-group designs, respectively.

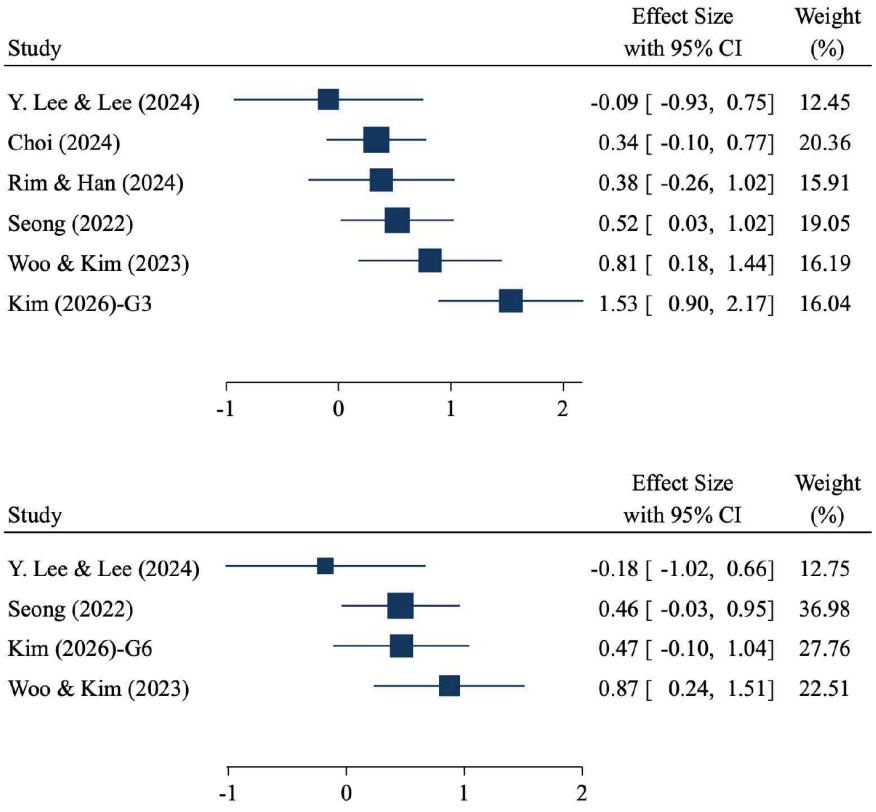
<Table 1> Mean Effect Size Estimates by Design and Outcome Domain

Stratum	***k***	***n***	***g***	***SE***	**95% CI**	***p***
Within–Cognitive	12	35	0.68	0.10	[0.47, 0.88]	< .001
Within–Affective	6	16	0.59	0.14	[0.31, 0.86]	< .001
Between–Cognitive	6	13	0.53	0.17	[0.20, 0.86]	.002
Between–Affective	4	13	0.41	0.17	[0.08, 0.74]	.014

Note. *k* = number of independent samples; *n* = number of effect sizes. All estimates are adjusted means (predictive margins) derived from two-level models including a domain indicator, estimated separately for within- and between-group effect sizes.



<Figure 3> Forest Plots for the Within-Group Samples: Cognitive (upper) and Affective (lower) Domains



<Figure 4> Forest Plots for the Between-Group Samples: Cognitive (upper) and Affective (lower) Domains

Two patterns in these estimates deserve emphasis. First, within each design, cognitive effects exceeded affective effects, at 0.68 versus 0.59 in single-group comparisons and 0.53 versus 0.41 in comparison-group designs. This asymmetry should nonetheless be interpreted as a descriptive tendency rather than a statistically established difference; the cognitive–affective contrast did not reach significance in either model (within: $b = 0.09$, $SE = 0.17$, $p = .620$; between: $b = 0.12$, $SE = 0.24$, $p = .623$), given the modest number of samples. Second, because single-group pre–post designs conflate treatment effects with maturation, regular instruction, and retesting (Shadish et al., 2002), the between-group estimates provide the more conservative and causally more defensible benchmark; that these remained significantly positive, at 0.53 for the cognitive and 0.41 for the affective domain, suggests that the observed gains are not merely artifacts of these uncontrolled features of single-group designs but hold even when learners are compared with control classmates.

2. Moderation by Grade Level (RQ#2-1)

Moderator analyses were conducted in the within-cognitive stratum by entering grade level, teacher intervention, and duration simultaneously as predictors ($k = 11$, $n = 34$; one sample spanning Grades 3–6 was excluded because it could not be assigned to a single grade-level category). The within-affective stratum was subject to the same exclusion, leaving five

samples and 12 effect sizes, and the corresponding moderator model failed to converge; moderation results are therefore reported for the cognitive domain only. Table 2 summarizes the estimates.

<Table 2> Moderator Analysis for the Within-Cognitive Stratum (k = 11, n = 34)

Moderator	***b***	***SE***	***z***	***p***	**Adjusted *g***
Grade level (6th vs. 3–5)	0.60	0.30	1.99	.047	6th: 1.12 [0.67, 1.57]; 3rd–5th: 0.52 [0.28, 0.77]
Teacher intervention (yes vs. no)	0.30	0.39	0.77	.441	—
Duration (semester-long vs. shorter)	−0.07	0.26	−0.28	.782	—

Note. Adjusted *g* values are predictive margins holding the other moderators at their observed values; brackets indicate 95% CIs.

Grade level significantly moderated the effect of AI PengTalk: holding teacher intervention and duration constant, sixth graders exhibited a large adjusted mean effect (g = 1.12, 95% CI [0.67, 1.57]), whereas third-to-fifth graders showed a medium effect (g = 0.52, 95% CI [0.28, 0.77]; contrast b = 0.60, p = .047). This pattern is consistent with the expectation that older learners' greater cognitive maturity allows them to exploit the structured rehearsal affordances of flipped learning and spaced practice more fully, producing pronounced pre-to-post growth.

This finding, however, requires cautious interpretation. The grade-level moderation could be tested only in the single-group (within) design; the comparison-group (between-cognitive) stratum contained too few samples (six) to test whether the sixth-grade advantage survives a causal design. Accordingly, the large sixth-grade estimate is best read as a robust descriptive trend awaiting causal confirmation, rather than as evidence that AI PengTalk causes larger gains for older learners.

3. Moderation by Teacher Intervention and Duration (RQ#2-2, RQ#2-3)

Neither teacher intervention nor duration reached statistical significance as a moderator, but both displayed directional trends consistent with the qualitatively derived expectations. Implementations involving active teacher intervention were associated with larger effect sizes than purely self-directed uses (b = 0.30, p = .441), and semester-long implementations were associated with slightly smaller effect sizes than shorter ones (b = −0.07, p = .782). Given the limited number of samples per condition, these analyses were underpowered, and the non-significant results should be understood as an absence of evidence rather than evidence of absence; the direction of the trends is nonetheless treated as descriptive and hypothesis-generating when taken up in the Discussion.

V. Discussion

This meta-analysis quantified the educational effects of AI PengTalk on Korean primary English learners based on 77 effect sizes from 28 independent samples across 16 studies, and examined whether the pedagogical mechanisms proposed in the preceding qualitative synthesis (Lee, 2026) are reflected in the quantitative record. Three findings warrant extended discussion.

1. AI PengTalk as a Cognitive Rehearsal Tool: A Descriptive Cognitive Advantage Across Designs

The most consistent finding is the within-design asymmetry between domains: medium-sized cognitive effects (g = 0.53–0.68) versus smaller affective effects (g = 0.41–0.59), under both single-group and comparison-group designs, although, as noted, the domain contrast did not reach significance in either design and should therefore be read as a descriptive tendency. With only 18 within-group and 10 between-group samples contributing to the domain contrasts, moreover, these tests were underpowered, so the absence of significance cannot be taken as evidence of equivalence any more than the observed gaps can be taken as established differences. This asymmetry aligns closely with the qualitative conclusion that AI PengTalk works best as a ‘cognitive rehearsal tool’ within the regular curriculum. The tool’s core affordances (curriculum-aligned expression practice, unlimited repetition without evaluation pressure, and systematically scheduled review) map directly onto the conditions for pushed output (Swain, 1985) and distributed practice (Kang et al., 2014; Nakata, 2015), and the aggregated evidence indicates that these affordances yield dependable gains in vocabulary, pronunciation, phonics, and fluency. Notably, these estimates hold up well against the broader meta-analytic evidence when compared design by design. The between-group estimate (g = 0.53) exceeds the overall treatment effect of 0.39 reported for AI-guided individualized language learning relative to business-as-usual conditions (H. Lee & Lee, 2024), and the within-group cognitive estimate (g = 0.68) is nearly identical to the mean effect of 0.69 reported in a domestic meta-analysis of chatbot-assisted Korean EFL instruction (J. Y. Lee & Hwang, 2022), though it remains well below the within-group benchmark of 1.18 (H. Lee & Lee, 2024). Taken together, these comparisons indicate that a government-built, curriculum-aligned tool for young learners performs at least on par with the broader evidence base in the causally stronger between-group comparisons, despite its documented technical constraints.

The comparatively modest affective effects, which were small in the causally stronger between-group design (g = 0.41), are surprising at first glance, given that anxiety reduction and motivation enhancement were among the most frequently documented affordances in the qualitative literature (25 of 36 studies; Lee, 2026). Two complementary interpretations are

available. Statistically, the affective domain in this dataset pools conceptually heterogeneous subvariables, including attitude, interest, confidence, motivation (further subdivided into intrinsic, extrinsic, integrative, and instrumental types), self-efficacy, and self-directedness, whose effects likely differ in magnitude and possibly in direction (Lee et al., 2023); pooling them plausibly diluted the mean estimate. Indeed, some primary studies reported negative shifts on specific attitude subscales (e.g., attitudes toward AI's social influence) alongside positive shifts on motivation subscales, a pattern invisible in the aggregate. Read in an exploratory light against the Paradox of AI Scaffolding, the subvariables most closely tied to the tool's affective scaffolding, namely interest and confidence, tended to show the clearest positive shifts, whereas attitudes directed at the AI itself and self-regulatory constructs such as self-directedness were flat or negative, suggesting that gamified engagement rises faster than learners' evaluative stance toward the technology. Qualitatively, the fading novelty dynamic offers a substantive explanation. Perception studies conducted soon after the tool's launch conceptualized learners' initial fascination as a temporary novelty effect (W. Choi, 2021) and described interest waning with repeated use (G. E. Jeong et al., 2021; Seong & Lee, 2021), and declining engagement over an extended implementation period has since been documented (Y. Choi, 2024). Initial enthusiasm driven by the character and gamification thus subsides as the mechanical limits of the interaction become apparent, capping affective growth even where cognitive rehearsal continues to pay off (Lee, 2026). Both interpretations converge on the same implication: affective outcomes should be meta-analyzed at the subvariable level once the primary literature accumulates sufficiently.

2. The Sixth-Grade Advantage: Promise and Caveats

The grade-level moderation, with sixth graders at 1.12 versus 0.52 for third-to-fifth graders in single-group cognitive comparisons ($p = .047$), provides the first quantitative footprint of the learner-level differentiation proposed in the Paradox of AI Scaffolding. The promising side of this finding is pedagogically encouraging: it suggests that when higher cognitive maturity meets structured implementation models such as flipped learning and spaced practice (Kang et al., 2014; Lee, 2026; Nakata, 2015), AI PengTalk can drive large pre-to-post growth, presumably because older learners can self-regulate the rehearsal cycle that the tool affords (Dent & Koenka, 2016; Ha et al., 2023; Zimmerman, 2002). In this respect, the finding qualifies the blanket concern that the tool serves only beginners.

The caveats are twofold. Methodologically, the moderation emerged only in designs without control groups and could not be verified in the underpowered between-group stratum, so novelty, maturation, and instructional co-occurrence cannot be ruled out as contributors to the sixth-grade estimate; nor can grade-related differences in out-of-school English exposure,

which single-group designs are inherently unable to control: nationally, elementary students' participation in English private tutoring rises from about 36% in Grade 1 to 64% in Grade 5 (Statistics Korea, 2025, pp. 24, 27), so sixth graders enter the classroom with the longest cumulative exposure to such tutoring. Substantively, the qualitative record cautions against equating large short-term cognitive growth with sustainable engagement for advanced learners: precisely the learners who gain the most measurably are those documented to experience ceiling effects and boredom with the tool's evaluative feedback (Lee, 2026). The quantitative and qualitative records are thus not in tension but in dialogue: large cognitive gains and motivational attrition can coexist for the same learners, which is exactly the profile that hybrid scaffolding is meant to address.

3. Teacher Intervention and Duration of Use: Directional Trends Awaiting Verification

The moderator analyses for teacher intervention and duration did not reach significance, yet their directions (larger effects with teacher intervention, and slightly smaller effects in semester-long than in shorter implementations) are consistent with the two central claims of the qualitative synthesis (Lee, 2026): that unmediated AI use is insufficient, and that a novelty effect inflates early outcomes (W. Choi, 2021; Y. Choi, 2024). The teacher-intervention trend converges with the hybrid scaffolding account (Lee, 2026), in which teachers resolve ASR failures in real time, supply the corrective feedback the algorithm cannot, and sustain communicative motivation through human interaction. The duration trend, meanwhile, suggests that the durability of the measured gains should be interpreted with caution: most primary studies employed single posttests within one semester, so the aggregated estimates partly reflect the early, novelty-boosted phase of implementation. Together, these trends shift the primary question facing the field from whether AI PengTalk works on average to which human-orchestrated conditions allow its effects to persist beyond the novelty window, an inherently longitudinal question that the current evidence base cannot yet answer.

VI. Conclusion

This two-level meta-analysis of 16 studies, 28 independent samples, and 77 effect sizes yields three conclusions. First, AI PengTalk is effective in promoting cognitive achievement, producing consistent medium-sized gains in primary learners' vocabulary, pronunciation, phonics, and speaking performance across research designs. Second, it is a meaningful but comparatively modest affective tool, producing positive effects that remain descriptively smaller than the cognitive gains within each design and whose magnitude is likely capped by the pooling of heterogeneous affective subvariables and by fading novelty. Third, the moderation patterns (a sixth-grade advantage that was large in magnitude but verifiable only in

single-group designs, together with non-significant trends favoring teacher intervention and shorter implementations) are directionally consistent with the Paradox of AI Scaffolding and position teachers' hybrid scaffolding, which couples technology-generated learning data with attentive human mediation, as the leading working hypothesis for sustaining the tool's benefits across learners and over time, pending verification in adequately powered comparative designs.

Several limitations must be acknowledged, each pointing to a concrete direction for future research. First, the between-group evidence base remains thin (six cognitive and four affective samples), which precluded causal verification of the grade-level moderation; as comparison-group studies accumulate, the moderation analyses reported here should be re-estimated within causally stronger designs. Second, the affective domain pooled broad and heterogeneous subvariables, limiting the precision of its aggregate estimate; future syntheses should disaggregate affective outcomes (e.g., anxiety, motivation types, self-efficacy) once the primary literature permits. Relatedly, because the cognitive and affective effect sizes of the same learner group were treated as separate samples, the dependence between the two domains within a learner group was not modeled, which may render the precision of the cognitive–affective comparison slightly optimistic. Third, the evidence base consists predominantly of short-term implementations with single posttests, making it difficult to disentangle genuine acquisition from novelty effects; longitudinal studies spanning an academic year or more are needed to verify the sustainability of both cognitive and affective gains. Finally, because AI PengTalk was built expressly around the Korean national curriculum, the present estimates are tightly bound to that context; rather than limiting their value, this specificity positions the findings as a benchmark case of what a curriculum-aligned, nationally deployed AI speaking tool can achieve and what human mediation it requires. Refining and empirically testing hybrid scaffolding models that link AI-based individual practice with human-to-human communicative activity remains the most pressing agenda for both research and practice.

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An asterisk (*) denotes studies included in this meta-analysis.

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Appendix A. Characteristics of the 16 Primary Studies and the Sample Structure

Study	Grade	n (T/C)	Duration	TI	Cognitive outcomes	Affective outcomes	W-Cog	W-Aff	B-Cog	B-Aff
Cha (2023)	6	17/–	15 wks (semester)	N	Vocabulary	Attitude; interest; belief and expectation	1	3†	–	–
Choi & Lee (2023)	5	13/–	4 wks (12 sess.)	N	–	Interest; confidence; learning motivation	–	3	–	–
Y. Choi (2024)	4	41/42	3 wks (6 sess.)	Y	Word recognition (3 subskills); reading fluency (4 subskills)	–	7	–	7	–
Heo (2024)	4	19/–	One semester	Y	Speaking; speaking (self-assessed)	Affective composite	2†	1†	–	–
Jang & Kim (2022)	3	26/–	One semester	Y	Speaking fluency; speaking accuracy	Interest; confidence	2	2	–	–
J. W. Jeong (2025)	5–6	133/–	24 sess. (semester)	Y	–	Motivation (4 subscales); attitude toward AI (5 subscales)	–	9†	–	–
M. Kim (2026)	3	25/24	16 sess.	Y	Phonics	Interest; attitude; confidence	1†	–	1	3
M. Kim et al. (2023)	6	22/–	6 sess.	Y	Sentence repetition; dialogue practice (2); pronunciation subscores (4)	–	7	–	–	–
K. Lee (2022)	6	21/–	9 wks (18 sess.)	N	Vocabulary; sentence composition	–	2	–	–	–
Y. Lee & Lee (2024)	3–6	11/11	10 sess.	Y	Speaking	Interest; attitude; self-efficacy; self-directedness	1	4	1	4
Oh & Back (2022)	5	27/–	9 wks (18 sess.)	Y	Pronunciation; word speaking; sentence speaking; fluency	–	4	–	–	–
Rim & Han (2024)	5	20/20	4 wks (10 sess.)	Y	Vocabulary; pronunciation	–	2	–	2	–
S. Seong (2022)	5	32/33	6 wks (12 sess.)	Y	Vocabulary	Interest; motivation; confidence	1	3	1	3
Woo & Kim (2023)	5	21/21	2 wks	Y	Pronunciation	Interest; attitude; self-confidence	1	3	1	3
Yoo & Ahn (2024)	5	18/–	2 wks (6 sess.)	Y	Pronunciation ratings (AI & human raters; nativeness, intelligibility)	–	6	–	–	–
T. Yoon (2022)	6	45/–	One semester	Y	Speaking	Learning attitude	1	1	–	–

Note. *n* (T/C) = treatment/control group sizes; TI = teacher intervention; W-Cog through B-Aff = number of effect sizes contributed to each design-by-domain stratum; † = samples removed by the influential-case diagnostics. Before exclusion: 33 samples, 93 effect sizes, N = 1,141 (642 unique students across the 16 learner groups); after exclusion: 28 samples, 77 effect sizes, N = 928. Effect sizes for Yoo and Ahn (2024) were computed from utterance-level ratings; the participant count reflects the 18 learners.

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